CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (CL) Examination

December- January 2013

CL 702 Structural Dynamics & Earthquake Engineering

Date: 03.01.2014, Friday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 1893: 2002 Part-I, IS 4326: 1993, IS 13920: 1993 & IS 456: 2000 are permitted.

SECTION – I

Q – 1 (a) An unknown mass 'm' is attached to one end of the spring of stiffness 'k' having natural frequency of 6 cycles/sec. When 1 kg mass is attached in addition to the mass 'm' in the same spring, natural frequency of the system is lowered by 20 %. Determine the mass 'm' & stiffness 'k'. **[03]**

(b) A forced undamped vibration system consists of simply supported beam of 6 m span which supports a machine of weight 980 kN placed at the centre of the span. The machine exerts the harmonic force of 10 kN with the forcing frequency of 12 rad/sec. Calculate the depth of the beam required so that the maximum amplitude of vibration for the system is restricted to 5 mm considering the width of the beam as 0.5 m. Consider modulus of elasticity, $E = 2 \times 10^5$ MPa. **[04]**

Q – 2 Attempt **Any Two** of the followings: **[14]**

(a) An idealized single degree of freedom system consists of a water tank shaft of 2 m diameter & 100 mm wall thickness, which supports a container of 2000 kN weight at its top. The effective height of column shaft is 10 m. The damper of the system offers the resistance of 1000 N at the velocity of 2 m/sec. By neglecting the self weight of column, Calculate the damping ratio and state whether the system is under damped, over damped or critically damped. Also calculate the natural period & damped frequency. Consider modulus of elasticity, E = 25000 MPa. ($K = 3EI/L^3$)

(b) A single degree of freedom system consists of a steel column of diameter 'D' & height 3 m which supports the 3 m diameter slab. The slab carries the load intensity (all inclusive) of 4 kN/m^2. The system is subjected to a harmonic excitation of 500 sin(50)t N. Calculate what should be the diameter 'D' of the column so that the possible range of resonance can be avoided. Consider modulus of elasticity, $E = 2 \times 10^5$ MPa.

(c) A cable attached to the column applies the lateral force of 20 kN. Due to this, the top of column displaces by 5 cm. The cable is suddenly cut and the resulting vibrating motion is recorded. At the end of 5 complete cycles, the time is 5 sec. and the amplitude reduces to 2 cm. Calculate (i) natural period, (ii) damping ratio, (iii) damping coefficient, (iv) no. of cycles required for the displacement to reduce to 0.2 cm.

Q – 3 A four storied building has lumped floor weights from bottom to top as 200 kN, 250 kN, 250 kN & 100 kN with storey stiffness of 500,000 N/m for all stories. From the free vibration analysis, the natural frequencies and corresponding mode shape coefficients are obtained as follows: [14]

ω_1 = 1.80 rad/sec, ω_2 = 5.32 rad/sec, ω_3 = 8.04 rad/sec and ω_4 = 9.14 rad/sec

$\{\phi_1\} = \{\phi_{11}, \phi_{21}, \phi_{31}, \phi_{41}\} = \{0.0018, 0.0033, 0.0043, 0.0046\}$

$\{\phi_2\} = \{\phi_{12}, \phi_{22}, \phi_{32}, \phi_{42}\} = \{-0.0041, -0.0036, 0.0021, 0.0048\}$

$\{\phi_3\} = \{\phi_{13}, \phi_{23}, \phi_{33}, \phi_{43}\} = \{0.0051, -0.0030, -0.0014, 0.0047\}$

$\{\phi_4\} = \{\phi_{14}, \phi_{24}, \phi_{34}, \phi_{44}\} = \{0.0020, -0.0027, 0.0039, -0.0059\}$

Consider the building as ordinary residential building with ordinary RC moment-resisting frame (OMRF) proposed on medium soil at Changa. Calculate the design lateral force at each floor in each mode. Also calculate the storey shear force considering participation of all modes using modal combination method (SRSS or CQC).

OR

Q – 3 (a) A two storied building has lumped floor weights from bottom to top as 95 kN & 80 kN with storey stiffness of 5 x 10^5 N/m and 4 x 10^5 N/m respectively. Perform the free vibration analysis & determine natural frequencies and corresponding mode shape coefficients. Also sketch the mode shapes. [07]

(b) A three storied building has lumped floor mass from bottom to top as 7000 kg, 7000 kg and 3000 kg with storey stiffness of 15,000 N/m at all stories. From the free vibration analysis, the natural frequencies and corresponding mode shape coefficients are obtained as follows: [07]

ω_1 = 0.77 rad/sec, ω_2 = 2.12 rad/sec, ω_3 = 2.91 rad/sec

$\{\phi_1\} = \{\phi_{11}, \phi_{21}, \phi_{31}\} = \{0.0051, 0.0087, 0.0099\}$

$\{\phi_2\} = \{\phi_{12}, \phi_{22}, \phi_{32}\} = \{0.0100, -0.0010, -0.0099\}$

$\{\phi_3\} = \{\phi_{13}, \phi_{23}, \phi_{33}\} = \{-0.0042, 0.0081, -0.0117\}$

Derive the Rayleigh damping matrix such that the damping ratio is 5 % for the 1st and 2nd modes. Also calculate the damping ratio for the 3rd mode.

SECTION – II

Q-4 Answers the followings:

1 Damages due to earthquake are always allowed in the structures. Justify this statement. [01]

2 RC buildings are design as per "Strong-column weak beam" design method. Give [01]

proper reason of this statement.

3 Is magnitude of earthquake same as the intensity? Give proper reason of this statement. [01]

4 Simplicity and symmetry is the key to making a building earthquake resistance. Explain the concept with the help of examples. [02]

5 What is a non-structure? How does it affect the overall behavior of the building? [02]

Q-5(a) Discuss how soil and structure interact during an earthquake? [04]

(b) As an engineer give key point on Bhuj 2001 earthquake. [05]

(c) If a column of size 300 mm × 530 mm is having the reinforcement of 1.6 % of the c/s area. Detail the longitudinal reinforcement of the column satisfying all criteria of IS 13920:1993 along with the neat sketch of longitudinal section. Use M 20 & Fe 415. [05]

OR

Q-5(a) State the soil conditions under which liquefaction can occur. [04]

(b) What is base isolation? Explain any one base isolation system. [05]

(c) Explain the concept of ductile detailing and how can you improve ductility of structures? [05]

Q-6 Attempt **Any Two** of the followings: [14]

The plan and elevation of building are shown in **Figure 1**. Assume data for the building as follows:

Live load data:

LL on roof = 1.0 kN/m^2 (for seismic calculation = 0)

LL on floor = 1.0 kN/m^2

Dead load data:

Thickness of floor and roof slab = 120 mm

Weight of slab = 3 kN/m^2 (Assuming weight density of concrete = 25 kN/m^3)

Thickness of wall = 250 mm

Weight of wall = 5 kN/m^2 (Assuming weight density of masonry = 20 kN/m^3)

Seismic data:

Seismic zone = V

Importance factor = 1

Response reduction factor = 3

Medium type soil strata

Direction of seismic force = E-W direction

(a) Design lateral load with lateral force distribution diagram.

(b) Relative rigidity of exterior shear walls (North and South).

(c) Torsional shear forces in North and South shear walls (If relative rigidity of each shear wall is 0.5).

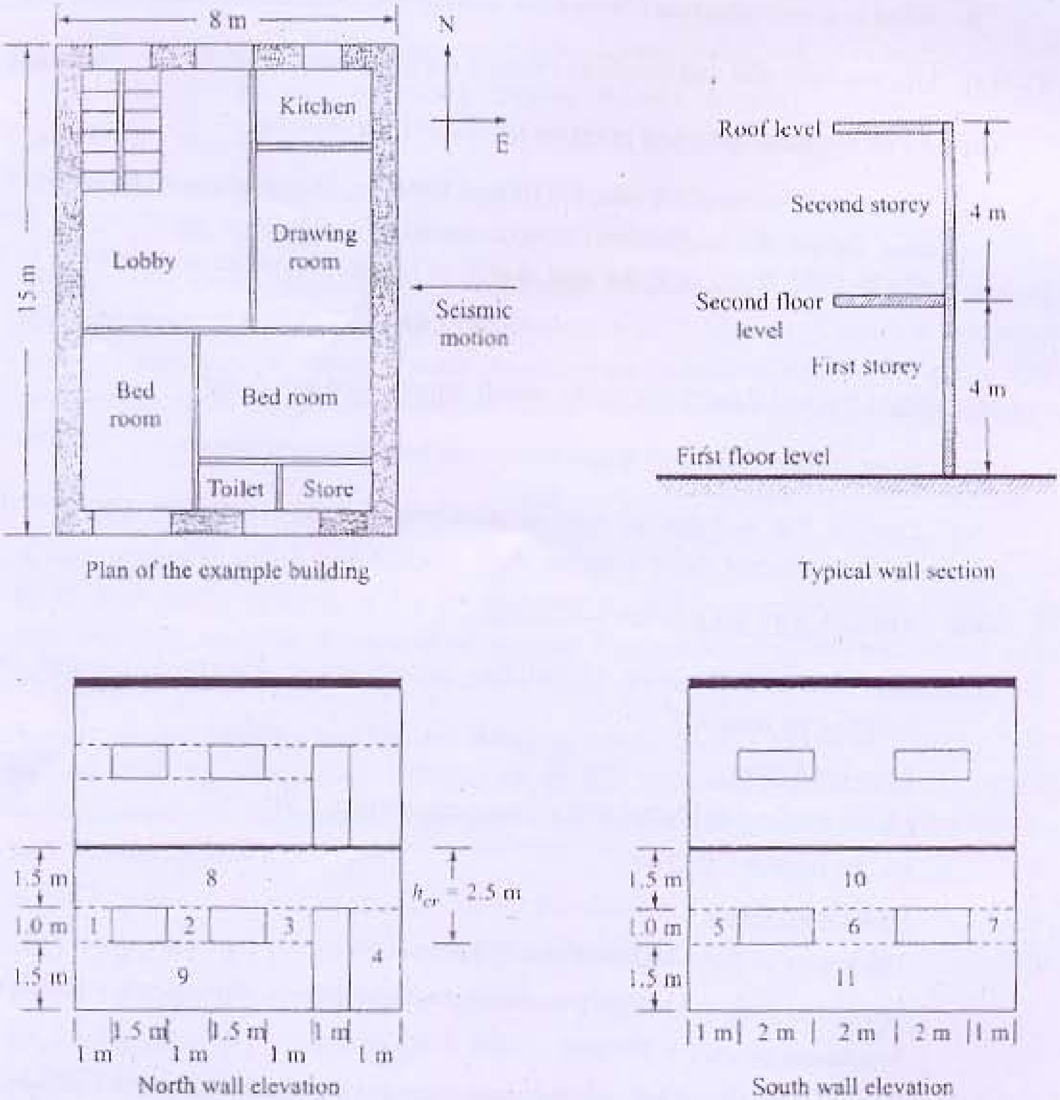

Figure 1